Analysis of Lap Times in Formula-1 Motorsport due to Regulation Changes Using Polynomial Regression

,Karthikeyan K[1] Thejineaswar Guhan[2]

[1]*Professor, Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology Vellore*
[2] *School of Information Technology and Engineering (SITE), Vellore Institute of Technology Vellore*
Email: k.karthikeyan@vit.ac.in

Abstract: *In the motorsport of Formula 1 engine regulation changes are quite common and when a change occurs the sport evolves by adapting to the new regulations. However, this transition from one regulation to another are not always uniform. So, we study how an engine regulation change affects the performance of the car by comparing the lap times and predict future performance due to new regulations passed using polynomial regression. For this study, we compare data spanning over 3 engine eras-V10, V8 and V6- during a 25 year-time period. As mentioned, the performance of these engines is analyzed and predicted using the lap times: it is the time taken in seconds to cover the entire track once. A lap time which is relatively smaller in a specific track, indicates better performance of the car in that track.*

1. Introduction

Formula 1 or F1 is the pinnacle of single seater racing motorsport and it is one of the most competitive ground, where milliseconds make huge differences. In F1, 20 drivers part of 10 teams compete in an approximate 20 rounds all held around the world. Each race lasts with a maximum of 2 hours covering almost 300km in distance. The driver securing the most points throughout the season is crowned as Driver's World Champion and the team with the most points is crowned as the Constructors' World champion. These points are awarded on each race on the basis of performance: with the highest one can score is 25 by winning the race and the lowest to be 1 point by securing 10th position. Started in 1950, the sport has continually evolved to what it is today. Formula 1's heart throbbing excitement and speed has captivated 1.922 billion audience on TV cumulative for the year 2019. Formula 1 is the dream of every karting and single seater driver to race for, hence the competition amongst the seats are also high due to the very less vacancies in seats.

As mentioned, the car is a single seater which means that the car only accommodates one person. As a result, teams have two drivers driving two cars. These cars can no way be compared with Sports car. These cars are engineered in a way which makes them capable of reaching speeds over 300 km/hour on straight sections of roads while also acing through tight and tough corners. To extract the maximum out of the car drivers, push beyond the limits which sometimes lead to accidents. Earlier there use to be fatal accidents which also have led to deaths. Introduction of safety became ever the more crucial and now the cars are safer than they ever could be.

Before a race starts all racers go through a phase or round called the qualifying. The qualifying is divided into three parts where five are eliminated in the first 2 rounds and the final 10 compete for the pole position or the first place of the starting grid. The fastest person around the track grabs the pole position. An example of this competitiveness is the season finale of 1997 held in Circuito de Permenante de Jerez in Spain. Michael

Schumacher and Jilles Villeneuve were competing for the championship, where Michael Schumacher was 3 points short of Villeneuve in the championship. Not only them but also Heinz-Harald Frentzen cloaked in the same time of. However, Villeneuve was given in pole as he was the first to put up that lap time.

New regulations are passed quite often, however these regulations are to do with new chassis structure, aerodynamics and etc. New engine regulations are passed however not as often as the normal regulation change. These changes include changing from V6 to V8. These engines are far more different and as a result the performance of the car changes. These changes as a result affect the lap time. The lap time in a way of measure of the car's performance.

A lap time is the time taken by a car to complete one lap in a circuit. The faster the car, the lower the lap time. This means that lap times indicate the strengths and weaknesses of the cars as well. These can include cornering abilities, straight line speed, acceleration and many more. In each tracks lap times are divided into sectors which make them much more favorable to analyze their cars.

In this article we look at how an engine regulation change affects the performance of the car. As mentioned above we look at the performance by comparing lap times. As a result, we compare data spanning over 3 engine eras-V10, V8 and V6- during a 25 year-time period. We analyzed the performance of these engines using the lap times.

1. Literature Review

Gibbs Y. Kanyongo.et.al.[1] establish the relationship between home environment and reading achievement using regression analysis. Alireza Mohebbi.et.al.[2] studied the relationship between the article title length and references and self-generated ideas using Linear Regression**.** Xu ZHANG.et.al.[3] discover the best possible relation between height and weight and Linear Regression is utilised for the same. Tea Baldigara.et.al.[4] analyse the annual wages for the past 20 years and derive a second-degree polynomial regression relation. M.K.N. Arshad.et.al.[5] focus on finding the probability of the outage of the line occurring and to assess the collapse of voltage due to the same using polynomial regression in their study. This method is used because of the uncertainty of finding such values using other methods.

Karthikeyan.K.et.al.[6] in their study predicted soya-bean prices in USA using multi-linear regression method. This study is focused on components which play an important role in the prediction however were never utilised before. Sartaj Singh Wariah.et.al.[7] employ linear and polynomial regression to inspect the relationship between the India as a country's growth with the respect to the presence of further studies. Oliver Budzinski.et.al.[8] analyse enterprise side of Formula 1 while also analysing the competitive balance, which simply looks at the power equality of the teams in Formula 1. A higher competitive balance indicates a grid which is fair and competitive in nature. This aspect is analysed using various factors within the sport. Shubham R. Ugle.et.al.[9], examine safety in Formula 1. They do this by covering the technical and precaution progress in safety. The study goes in depth covering the flags used in the races to equipment used by the drivers which increase the safety in general. Triya Nanalal Vadgama.et.al.[10] investigate the aerodynamic structure of a Formula 1 car. They look into the structure and components of the car which lead the car to have higher downforce which leads to a car which is faster around the corners in general.

2. Data sources and General Information

The data taken for this study is taken from the official Archives of Formula 1 website (https://www.formula1.com/en/results.html/2019/races/1000/australia/race-result.html) Sample periods for the given study is from 1995 to 2019. During this period there have been 8 circuits which have remained constant. There are some tracks which started in between this period and we have not considered them as well.

Some other information regarding the specifications of the car(http://www.formula1-dictionary.net/engine_rule_changes_history.html) was taken form formula 1 dictionary. The development of the wheel bearing system of the Renault was obtained from Wikipedia(https://en.wikipedia.org/wiki/Renault_R26)

Information about the type of track was obtained from Scuderia Alpha Tauri's Official website.(https://www.scuderiaalphatauri.com/en/f1-tracks/)

Information in the introduction about the close qualifying of Michael Schumacher, Jilles Villeneuve and Heinz-Harald Frentzen was obtained from Racefans.net (https://www.racefans.net/2017/05/25/1997-spanish-grand-prix-flashback/)

We are considering the data from the qualifying section of the racing weekend. Qualifying determines the track position of the car on race day hence it is crucial as some tracks have less opportunities to overtake. Hence, teams use the best setup and low fuel for the least weight. In a way through this we find the best performance of the car.

Of this 8 circuits we have selected 3 circuits which show us a clear trend. Some of the values in the other circuits are affected by sudden extreme values and these circuits have different needs from the car. There are two factors which we cover: downforce and straight-line speed.

Downforce, is the ability to keep the car as close to the ground. As mentioned above these cars blast through circuits with high speeds and because these cars travel at such a speed the cars the grip of the cars decreases automatically. When the grip decreases, the lap time decreases hence never letting the car show its true potential. Downforce is achieved using components which are made using the principles of aerodynamics. The aerodynamics in the plane keeps it above the ground. Here similar principles are used in the opposite direction in order to keep the car as close to the ground.

Straight-line speed as the name suggest is the maximum speed attained by the car on the straight lines of the car. Some circuits tend have more Straight-line stretches than the other therefore making it very key. Acceleration of the car also plays a role in order to attain a good straight- line speed because the car needs to spend time in the maximum speed for the maximum time on the straight line in order to cover the distance faster.

In a way we cover major aspects of cars by analyzing factors which bring about these performance changes

Below are the selected circuits with respect to their country:

1. Brazil-A high downforce Circuit
2. Spain-A high-medium downforce Circuit
3. Canada-A low downforce Circuit

We talk about different eras, to be more specific these are the eras:

1. Era 1: 1995-2005, V10 Engine used
2. Era 2:2006-2013, V8 Engine Used
3. Era 3:2014-2019, V6 Engine along with the power unit being used (Current Specification of engine used)

4. Methodology

Regression analysis is used to model the relationship between a response variable and one or more predictor variables. Regression analysis helps one understand how the value of the dependent variable (or 'criterion variable' or Y) changes when any one of the independent variables (X) is varied, while the other independent variables are held fixed. Regression analysis is widely used for prediction and forecasting, which is our end goal to predict the future of the motorsport due to new engine regulations.

The general multilinear polynomial regression is

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4 + \cdots + \beta_r x^r$$

In most situations we end up getting a non-linear relationship hence we have opted to go for a maximum of 4-degree polynomial regression.

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4$$

The data extracted for this analysis was in the format of minutes: seconds: milliseconds. Hence, the data was converted into the format of seconds. Then the data was fed into the R Software and using the software we fit the technique to the data. After this, we plot the fitted data and the result is the plots for the respective cases.

In this article, the time in each era is considered as independent variable and the lap times is considered as dependent variable. We analyze each era individually and also analyze the 25year time period all together to show the trend as well.

5. Result Analysis

5.1. Case 1: All Eras

We look at the trend and forecasting using 4- degree regression and the best line fit is found.

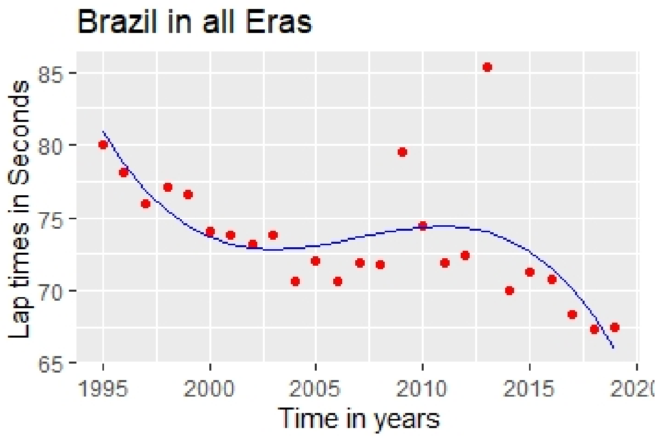

Figure 1. The trend of Brazil in all Eras

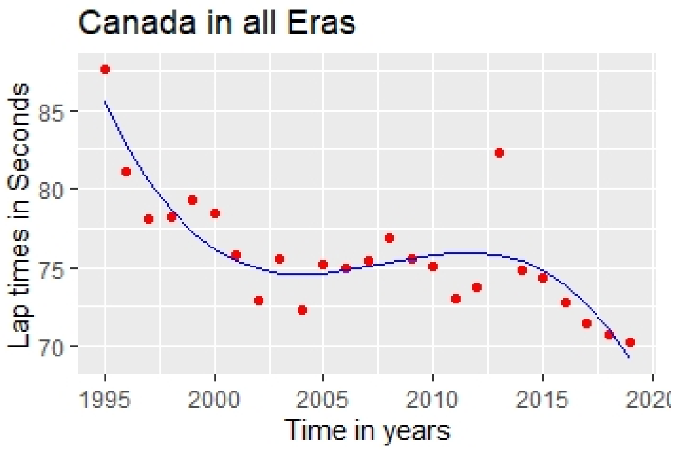

Figure 2. The trend of Canada in all Eras

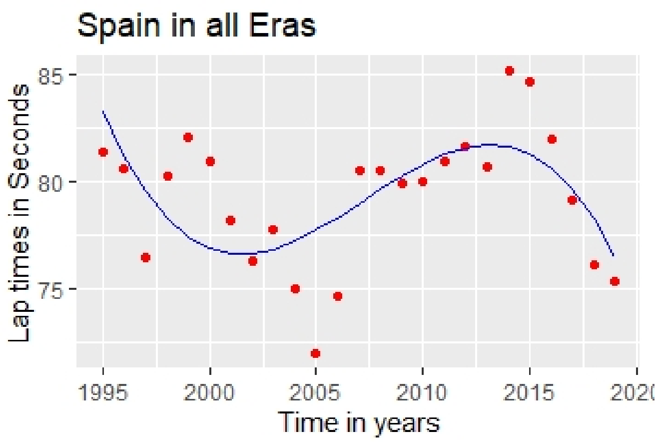

Figure 3. The trend of Spain in all Eras

The all in all trend shown by these three are somewhat similar. All of them point a downward trend, which indicates that the new engine specifications combined with the electrical components are faster which shows the power of development in the car. Not only the car, but the drivers have adapted themselves to tame these fast machines.

Given below are the regression models of all the four circuits. Here x indicates the number of years since 1995.

Brazil

$$y = -(8.077e - 07)x^4 + 19.52x^2 - 52230x + (3.931e + 07) \quad (1)$$

Spain

$$y = -(8.133e - 07)x^4 + 19.66x^2 - 52620x + (3.961e + 07) \quad (2)$$

Canada:

$$y = -(8.203e - 07)x^4 + 19.84x^2 - 53110x + (3.999e + 07) \quad (3)$$

Here we see that all the three figures have a similar shape, a horizontal ‘S’ shape. This signifies that the car has faced the performance of the car has been similar throughout all three types of tracks. We can say that there have been 2 eras which have seen better performance improvements and 1 era has seen a downgrade in the performance. Also, it can be seen that the equations are almost identical in nature. We will cover each era in detail further on

5.2 Case 2: Era 1

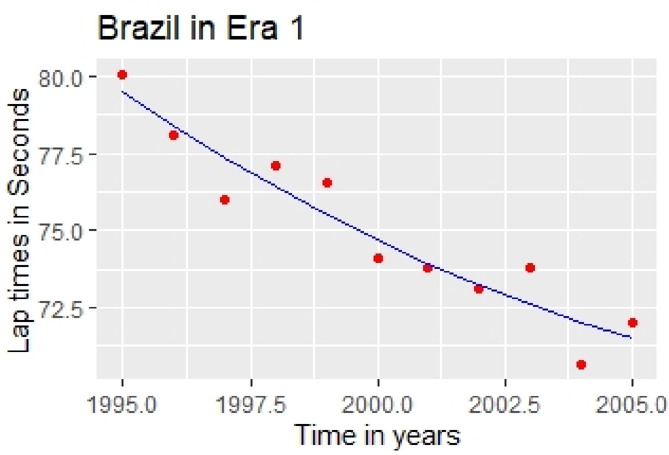

Figure 4. The trend of Brazil in Era 1

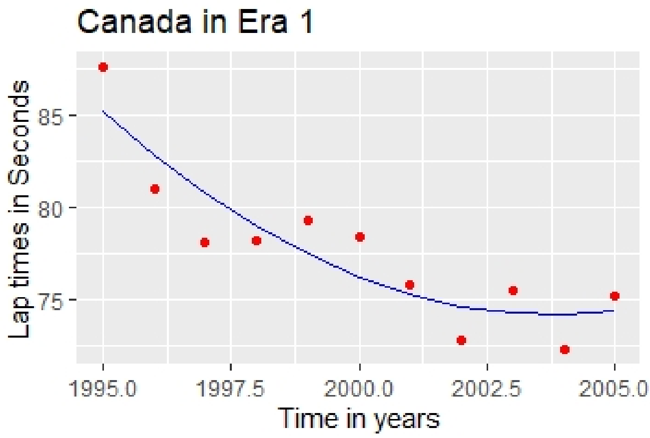

Figure 5. The trend of Canada in all Era 1

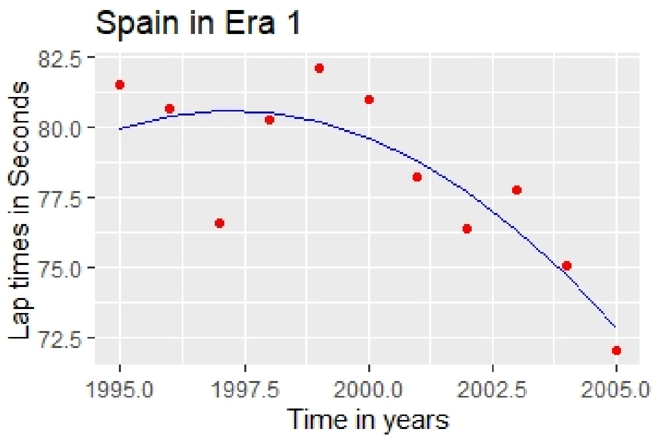

Figure 6. The trend of Spain in all Era 1

During the era 1, the engine was regulated to have a capacity of 5L. During this period the teams used a variety of engines at the start, ranging from V8 to V12. The V10s were giving the best result and as a result it became a preferred option. As a matter of fact, the 2005 season was restricted to 3 L V10 engines.

Below are the results

Brazil

$$y = 0.03297x^2 - 132.7\text{x} + 135500 \quad (4)$$

Spain

$$y = -0.1291x^2 + 5.6x - 51480 \quad (5)$$

Canada

$$y = 0.1435x^2 - 575.2x + 576400 \quad (6)$$

The era started off with a starting time off with a timing of 87.661 seconds in Canada to 75.217 seconds in Canada at 2005. This comes as teams had enough time to develop the perfect engines and also these engines were naturally aspirated meaning no usage of external devices to pump air in. The evolution is clear as the trend completely agrees with all the three circuits. To give a clarification all these data are taken from the fastest car during the qualifying session. In a way we find out the best of the development of that race in the season.

The improvement in Spain and Brazil also signify that the improvement in the cornering abilities of the car. Canada is a circuit which requires a low downforce setup. The higher the downforce the lower is the straight-line speed. However the car is faster round the corner. This conveys that both engine and aerodynamic developments were made. During this period a lot of developments were made. To give you an example of aerodynamic development, we can see the mass damper used by Renault in their F1 car front wing in 2005, this gave the car better stability over kerbs and through fast and slow corners.

The engine used during this era had a break horse power (bhp) range of 650 to 950. During the 2003 season the BMW P83 engine cleared the 900 bhp mark or 19200 revolutions per minute.

As shown Brazil(Figure 4) and Canada(Figure 5) clearly show a downward trend this indicates that the engines used during this era has developed which provides the car to produce about faster lap times. Also we can say that the lowest ever value is reached between the period 2004-05 which means that the development has been cumulative.

Similar to Brazil (Figure 4) and Canada(Figure 5), Spain(Figure 6) also has a downward trend. However it had reached a maximum value 1997. Although the trend in Brazil should agree with the one in Spain, unaccounted factors such as track temperature and wind may have affected the car's performance in general.

Considering the coefficients of x^2, we can say that our inferences are proved. In general, in a quadratic equation the value of x^2 affects the shape of the curve. When the value is smaller, considering in power of ten, the curve tends to be wider at the points of stationary points. A stationary point is the point on the curve where the gradient(slope) of the curve becomes zero and the direction of the curve changes as well. In this we look at the magnitude(signs ignored) of the coefficient of x^2, and we see that Brazil(0.003297) has the smallest coefficient and Canada(1.435) has the largest coefficient. Here Canada and Spain have reached their stationary point however Brazil is still in the process. If we consider slope, we see that Spain attains a much larger value when we start from the stationary point. Spain attained the stationary point at 1997-1997.5 with a lap time of approximately 80.5 and at 2005 the lap time is 72.6 seconds, which is 7.9 second decrease in a period of 7.5-8 years. If we consider Brazil, the stationary point has still not been reached and the rate of change from 1995 to 2005 is also 8 seconds but over a period of 10 years. This analysis shows that car improvements were better at low-medium downforce tracks than on a high downforce track.

5.3 Case 3: Era 2

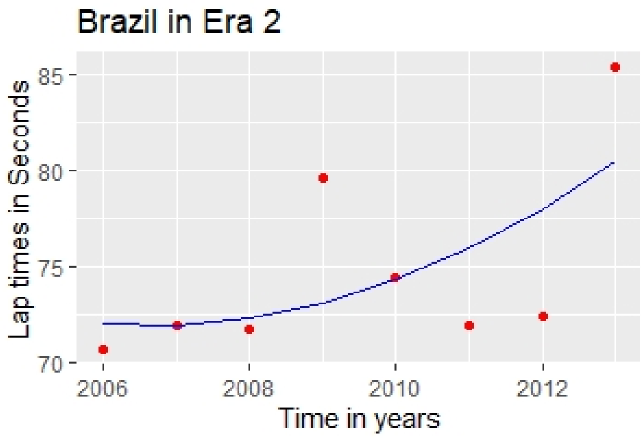

Figure 7. The trend of Brazil in all Era 2

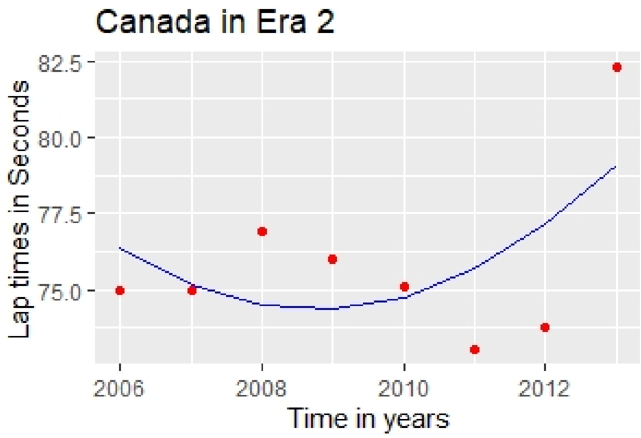

Figure 8. The trend of Canada in all Era 2

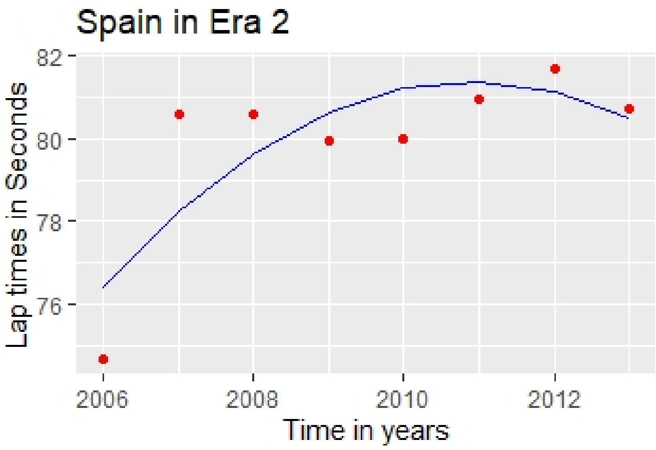

Figure 9. The trend of Spain in all Era 2

Brazil

$$y = 0.2115x^2 - 848.7\text{x} + 851600 \tag{7}$$

Spain

$$y = -0.2058x^2 + 827.7x - 832100 \tag{8}$$

Canada:

$$y = 0.2639x^2 - 1060x + 1065000 \tag{9}$$

Here we see a change of era, although the engine had changed the season started of fast with a 74.942 seconds. However, the trend remained constant as making new developments seemed difficult because more regulations were added. These regulations made the car go slow. During this era only Red Bull formula 1 team cracked the perfect car, the others were having a not so perfect car. We can say that during this period the timings fluctuated up and down which all in all contributed to an upward trend in terms of timing. However, the timing did go fast at occasions. There is one thing which we need to consider, the Canadian Grand Prix was not held in 2009 hence we have replaced that value with the average of the era 2 timings.

In 2006, the lap put in Hungary was 79.599 seconds. This time is faster than previous era. However, the timings never reached this height as the timings kept going up to 78s to 80s.

As said regulations regarding engine power was released which limited the pace of the car. The engine regulations were kept constants to drive development costs down, this led to no development during the period 2007-09. During the 2007 and 2008 season the power also looked by revolutions per minute(rpm) was kept constant at 19000. It was further decreased to 18000 rpm Also to consider is the fact that frequent engine changes could not be made. To be precise each driver had an allowance of 8 engines in a season from 2007. Later in 2008, it was introduced that one gearbox should be used for a minimum of 5 race weekends. Teams had to be frugal in terms of not overusing these components, which otherwise lead to penalties which would easily affect their race. Considering tracks such as Monaco where qualifying positions are key for victory. In 2008 the ECU or the Electronic Control unit was introduced which in a nutshell had the entire control of the car. This was a very new introduction which could be a factor why there is a performance dip.

As it can be seen the trend in almost all the three agree. By using Figure 8, we can clearly say that the engine power has been far lower than the previous era. This can be supported by the fact that engine power was kept constant throughout the first 3 years and as a result the engine development has not yielded better results after the constraint was removed. Moreover, the aerodynamics haven't been significant either. Aerodynamically demanding tracks such as Brazil and Spain have not yielded the desired results of a faster car.

In this case we see that the magnitude is almost the same with all values in the range of 0.2-0.3 seconds. Also to be considered is the fact that the all the cases have attained the stationary point and if we see the fact that in Brazil the in a period of 7 years we see an increase of approximately 6 seconds and in Canada in period of 5 years the lap time increases by 5 seconds. Brazil and Canada reached the stationary point early on however Spain reached the stationary point later on due to the negative sign on the coefficient of x^2. Here we see that in the first 5 years the lap time increases by 5 years. In general, we see that the timings increase by a second a year. As the slump attained in Spain was after having an increase first, naturally the direction of the curve is to change and that is what happens. In general, we can say that performance was not favorable in any type of track.

Case 4: Era 3

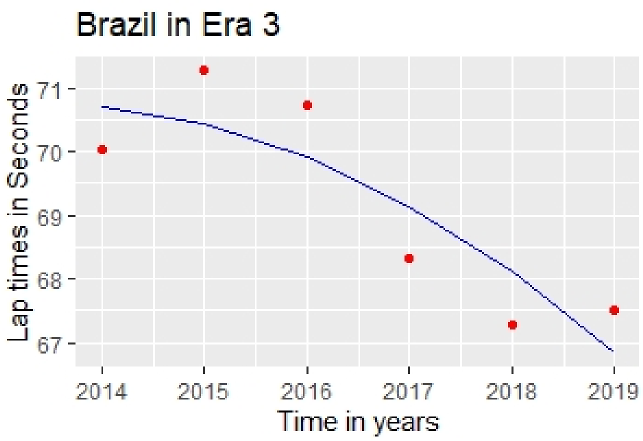

Figure 10. The trend of Brazil in all Era 3

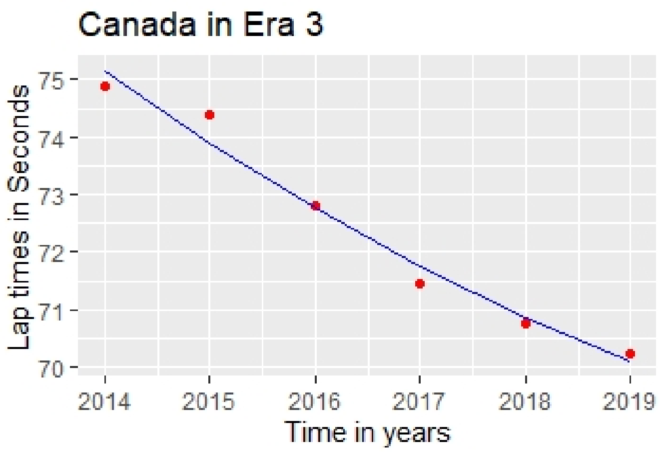

Figure 11. The trend of Canada in all Era 3

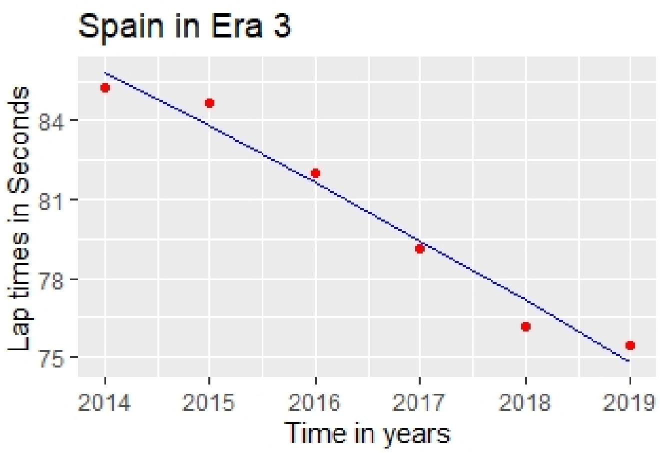

Figure 12. The trend of Spain in all Era 3

 ISSN: 0031-4773

Brazil

$$y = -0.1275x^2 + 513.4x - 516800 \quad (10)$$

Spain

$$y = -0.0436x^2 + 160.5x - 159600 \quad (11)$$

Canada:

$$y = 0.05945x^2 - 240.8x + 243800 \quad (12)$$

The start was a very rocky one. In fact, timings had reached the lowest ever. In 2014 the fastest time in Japan was 92.506 seconds, the lowest in last 10 years. Soon, this era of formula 1 started breaking records. Record of the fastest ever lap cloaked in 2005 in Italy was broken in 2018. Not only Italy, but many records were broken in various circuits.

This era started using hybrid technology to enhance the performance all in all. This era is still not over yet and 2020 being the last year we need to wait in order to look if this trend upholds or not.

Similar regulations as to the number of power units were passed in 2014. Drivers could use a maximum of 5 power units in a season. The power unit involves the engine and engine recovery systems and components which aid it to work. This regulation was further decreased to 3 power units per season and also the MGU-K was restricted to 2 per season. These restrictions were giving lower allowance than the previous era, which meant that the reliability became ever the more important.

Here, all the three figures completely agree that the performance has increased. We can say that the introduction of power unit has improved the performance of the car drastically. It can also be noticed that Aerodynamics have also been because all three circuits having somewhat of different needs converge to form an almost similar trend.

Here we see some different trends, by looking at the graph itself we see that Brazil and Spain are already past their stationary point. This indicates a positive relation of these timing to one aspect of the car which is downforce, downforce is crucial for Brazil and Spain. When considering Canada, we see it approaches the stationary point and this indicates that maximum engine power is being approached. So, we can say that the teams have extracted the maximum of the engine and the electrical components as some regulations are to be passed for the electrical components, so that slump is justified and the performance will naturally decrease for a few years to come.

5. Conclusion

All in all, the performance has increased throughout the engine era. We can also infer that it took time for the sport to develop to switch to different forms of performance boosters. The first two eras were naturally aspirated and the third one being turbo hybrid. In a way at one point of time the room for development had fairly decreased with Era 2 being the perfect example. We can fairly say that teams do adapt up to new regulations because they are informed in advance. However, making developments is no easy task. Also, these developments also show the technological advancements and also the various aspect of technicalities considered for updates.

Below are the values of the Multiple R-Squared values. These indicate the accuracy of the model. The higher the values the greater is the accuracy.

Table 1

Multiple R-Squared values			
	Brazil	Canada	Spain
All Eras	0.4827	0.7332	0.3957
Era 1	0.9004	0.8098	0.7032
Era 2	0.372	0.3181	0.6409
Era 3	0.75	0.9761	0.9693

Also, the new regulations set for 2022 bring about some changes to these components hence impacting the performance. Moreover, the budget cap set to be introduced in 2021, of $150 million is way lower than Ferrari and Mercedes budget of $400 million+. This could impact the development of new components. Although the changes to be made are minor, chances for new developments will be hampered. Probably we will never get to see quirky developments such as the DAS made by Mercedes F1 team. Although the field will be leveled the cars may never be as fast as before.

On the bright side we can also say that the engineers will find a way to improve their cars, like they recovered from the start of Era 3. All regulations made bring about a shockwave however this leads to new developments. Performance can be slow at the start however it will some way or other pick up later on. This growth will be restricted only if additional regulations other than budget is added. An example could be the fixed engine power throughout the start of Era 2.

Due to the current pandemic situation, the races have been occurring in tracks which have not been visited at all or have not been visited for some period of time. This change of tracks was caused because usual race hosts backed away to host the race. Hence we could only test our model across one circuit, Spain. The result is 76.512 seconds from the model which covers all eras and 74.770 seconds which covers only era 3 while the time achieved on track 75.584 seconds. We can say that our model has fairly predicted the time given the fact that in Spain there is a sudden decrease in timing in the last 3 years. Also, the timing achieved this year was actually slower than the previous years and the reason to this could be the increase in temperature. This happened as the race was rescheduled from May to August.

Regardless of all these factors, we believe that our model is capable of predicting the lap times in the upcoming seasons.

6. Acknowledgements

We acknowledge official Formula 1 website, Formula 1 dictionary website, Scuderia Alfa Tauri’ s official website and Racefans website for providing relevant information for research and analysis.

7. References

7.1 Journal Article

[1] Gibbs Y. Kanyongo, Janine Certo, Brown I. Launcelot," Using regression analysis to establish the relationship between home environment and reading achievement: A case of Zimbabwe", International Education Journal, 7(5), (2006), pp 632-641

[2] Alireza Mohebbi and Yousef Douzandegan,"Linear Regression Analysis of Title Word Count and Article Time Cited using R", Journal of Scientometric Research,2017,6,1,15-22, (2017) , pp. 15-22

[3]Xu ZHANG, Chao ZHAO , Hai-tao WANG, Fang ZHANG, Jing ZHAO and Gang WU,"A Regression Equation Model for Height and Weight Prediction", IOP Conf. Series: Journal of Physics: Conf. Series 1087, Issue 2, (2018).

[4] Tea Baldigara and Kristina Duvnjak, " Modelling Wages in Croatia Using a Second Order Polynomial", International Journal of Business Administration, Vol. 6, No. 6 , (2015), pp. 15-22

[5] M.K.N.Arshad , N.Aminudin , M.Marsedek , S.Z.M.Noor , R.H.Salimin , D.Johari, "Development of Transmission Line Failure Rate Model using Polynomial Regression", International Journal of Engineering & Technology, Vol 7, No 3.15, (2018)

[6]Karthikeyan.K, AkshayHarlalka, "Robust regression model for the prediction of Soybean crop prize based on various factors", International Journal of Emerging Technologies in Computational and Applied Sciences, Vol.8(3), March-May, (2014), pp. 258-263.

[7] Sartaj Singh Wariah and K Karthikeyan, "Trends in higher education and growth in India: A case study", Journal of Physics: Conference Series, Volume 1139, International Conference on Applied and Computational Mathematics 10 August (2018)

[8] Oliver Budzinski and Arne Feddersen," Measuring Competitive Balance in Formula One Racing", Ilmenau Economics Discussion Papers 121, (2019)

[9] Shubham R. Ugle1, Shweta D. Kate, Dhananjay R. Dolas," Formula One Safety: A Review", International Research Journal of Engineering and Technology, Volume 02, Issue 07, (2015)

[10] Triya Nanalal Vadgama, Mr. Arpit Patel, Dr. Dipali Thakkar, Design of Formula One Racing Car, International Journal of Engineering Research & Technology (IJERT), Vol. 4 Issue 04, (2015)